

The Model, Least Squares and the Normal Equations

A regression model is a claim about how the distribution of a response changes with a set of regressors, and the linear model is the simplest useful claim of that kind. It says that the mean of the response is a linear combination of known columns, that the errors around this mean have a common variance and are uncorrelated, and nothing more. This chapter sets the model up and draws out what follows from those assumptions alone. It starts with what the assumptions do and do not say, fits a straight line by calculus, and passes to the matrix form of the model, where polynomials, harmonics, indicator variables and interactions all turn out to be linear models. It then derives the least squares estimator and the normal equations, computes the mean and covariance of the estimator, estimates the error variance without bias, and asks what a fitted coefficient means. The last section surveys the models the rest of the book builds on this one. Throughout, the model matrix has full column rank. [Chapter 6](#) retells the same story geometrically and removes that condition.

What you need. [Chapter 1](#): rank (Proposition 1.2.3), inverses, nonnegative definite matrices (Proposition 1.7.5), traces (Theorem 1.6.1) and the derivatives of linear and quadratic forms (Theorem 1.11.1). [Chapter 2](#): means and covariances of linear transformations (Theorem 2.2.1), the mean and variance of a quadratic form (Theorem 2.3.1, Theorem 2.3.3) and the laws of total expectation and covariance (Proposition 2.5.4). Normality ([Chapter 3](#)) appears only in remarks.

5.-1.1 Roadmap

- [What a regression model claims](#)
- [Simple linear regression as a warm-up](#)
- [The multiple linear regression model](#)
- [Least squares and the normal equations](#)
- [Properties of the estimator under second-moment assumptions](#)
- [Estimating the error variance](#)
- [Interpreting coefficients](#)
- [A panorama of regression models to come](#)
- [Summary and notes](#)